9.2 Density Based on Area

Lesson Introduction

Benchmark Focus	MA.912.GR.4.4 Solve mathematical and real-world problems involving the areas of two-dimensional figures.	Learning Target	• I can solve real-world problems that involve density using area.

	Building On	Working Toward
Benchmark Coherence	MA.6.GR.2.2 Solve mathematical and real-world problems involving the areas of quadrilaterals and composite figures by decomposing them into triangles or rectangles. MA.7.GR.1.1 Apply formulas to find the areas of trapezoids, parallelograms, and rhombi. MA.7.GR.1.4 Explore and apply a formula to find the area of a circle to solve mathematical and real-world problems.	MA.912.GR.3.4 Use coordinate geometry to solve mathematical and real-world problems on the coordinate plane involving perimeters or areas of polygons.

Essential Questions	• What is the relationship between area, population, and population density? • When given the population density, how is the area calculated?
Lesson Narrative	This lesson guides students on finding the population density when given the area and/or the population. Students will engage with real-world data to find the population density when given the area and the population. This lesson also provides several other types of real-world application problems relating to density.

Component Summary	9.2.1 Warm-Up (~5 min.)	Choosing which car you would rather buy *(SE p. 59)*	9.2.3 Guided Instruction (10 min.)	Relating density and area *(SE p. 61)*
	9.2.2 Collaboration (15 min.)	Finding population density *(SE p. 60)*	9.2.4 Guided Practice (10 min.)	Solving contextual problems with density as it relates to area *(SE p. 63)*
	9.2.5 Wrap-Up (~5 min.)	Students summarize their learning *(SE p. 64)*		

	Glossary	Materials	Additional Preparation
Resources	<u>Key Terms:</u> • Population Density <u>Additional Glossary:</u> • Area • Density	• (Optional) Colored Pens/Pencils	• N/A

	Common Misconceptions	Things to Consider
Teacher Tips	• Students may improperly identify the definitions of area and density. • Students may mistakenly identify the area versus the population density when completing questions. • In 9.2.3 Guided Instruction question 3, students may confuse the fractions when converting the units to miles.	• This lesson does not relate the concept of density to the concept of volume, which will be discussed in Lesson 6 of Unit 9. • Consider letting students draw or use colored pens/pencils for the diagrams in the lesson.
	Literacy and Language Routines	Mathematical Thinking and Reasoning (MTR) Standard
	Notice and Wonder In the beginning of 9.2.2 Collaboration, students are given a list of the populations and the areas of the five largest cities by population in Florida. Allow students to do a Notice and Wonder here and gather responses from students as they make the connection between the population and its area.	**MA.K12.MTR.7.1: Real-World Contexts** This lesson is a perfect opportunity to apply mathematics to real-world concepts, as this lesson was written specifically using real-world concepts. Allow students to fully engage with the mathematics in this lesson.
Differentiate Instruction	9.2.2 Collaboration can be differentiated according to students' needs. Visuals would be helpful for students. Instead of having students write the answers in the workbook, have students record the answers on a visual such as a whiteboard, chart paper, or any other vertical non-permanent surface (VNPS). Students will be able to have choice in how their group's information is displayed.	

Lesson Notes:

9.2.1 Warm-Up: Would You Rather?

Component Overview: *Students begin by looking at two scenarios of cars to buy. This gives the students an opportunity to discuss mathematics and to justify their answers using mathematics.*

9.2.1 Warm-Up: Would You Rather?

1. Which car would you rather buy?

Car A	Car B
$19,000 12% depreciation per year when you upgrade the car after 5 years.	$25,000 16% depreciation per year when you upgrade the car after 5 years.

2. Explain your answer.

Answers will vary. I would rather buy Car B. Even though Car B costs more upfront, it will be worth $10,455.30 after 5 years whereas Car A will only be worth $10,026.91.

This question also provides an opportunity to connect a part of the lesson back to the Literacy and Language Routine for Notice and Wonder. If time allows, have students talk and verbalize their answers.

For an auditory approach to 9.2.1 Warm-Up, allow students to verbally share their answers with either a partner or the whole class.

9.2.2 Collaboration: Population Density

Component Overview: *Students will begin working with the tables shown, where the five largest cities by population are given as well as the area of each city. Through 9.2.2 Collaboration, students will complete a Notice and Wonder as well as collaborate with other classmates to discover the relationship between population and area to determine how to calculate population density. The data for this assignment is real-world data as of March 2021.*

9.2.2 Collaboration: Population Density

1. The five largest cities in Florida and their populations are shown.

City	Population
Jacksonville	890,467
Miami	454,279
Tampa	387,916
Orlando	280,832
St. Petersburg	261,338

The table shows the area of Florida's five largest cities.

City	Area (square miles)
Jacksonville	747
Miami	35.87
Tampa	113.41
Orlando	102.4
St. Petersburg	61.74

What Do You Notice About the Population and the Area?
Answers will vary. Highest population is not necessarily the same thing as population density – the city of Miami squeezes almost half a million people into 36 square miles.

2. Discuss with your partner your understanding of population density. Describe the population density for the city or town you live in.
Answers will vary. I think it means the amount of people per square mile – or how many people you fit into an area.

Is there a specific relationship between area and population?

For extra differentiation, allow students to choose whether to discuss verbally or discuss by writing down their answers in the workbook. See the Differentiate Instruction guidance section on page 2 for other ideas.

Students should be discussing the math here and should be given the opportunity to do so throughout this Collaboration. Be sure to tie in the Notice and Wonder Literacy and Language Routine as well as MA.K12.MTR.4.1.

9.2.2 Collaboration: Population Density (continued)

3. Make predictions about the population density for Florida's five largest cities.
Answers may vary.

Largest Population Density	Smallest Population Density
Miami	*Jacksonville*

4. Explain your thinking.
Answers will vary.
I think Miami has this highest population density – it's fitting so many people into an area that is quite small. I estimate that Jacksonville will have the lowest population density because even though it has the highest population, it also has the largest area by a factor of 7.

5. Ask a classmate for the predictions they made as well as their reasons. Record their cities and write your summary of their reason. *You are the only person who should write in the middle two columns of the table.* Have the person initial next to your summary stating that your summary was correct.

	City	My Summary of Their Reason	Initials
Largest Population Density		*Answers will vary*	
Smallest Population Density		*Answers will vary*	

To find population density, divide the population by the area.

$$\text{Population Density} = \frac{\text{population}}{\text{area}}$$

For a kinesthetic approach, divide the room into five sections (one for each city) and have students share their predictions by going to the city they think has the largest population density. Repeat by doing the smallest population density.

Each corner then chooses a spokesperson for their corner to provide rationale for their response.

Students can then choose to stay in their corner or reshuffle based on their arguments.

9.2.2 Collaboration: Population Density (continued)

6. Determine the population density, in persons per square mile $(\text{persons}/\text{mi}^2)$, of the five largest cities in Florida and record them in the table from largest to smallest. Round answers to the nearest whole number.

City	Population Density (persons/ mi.^2)
1. *Miami*	12,665
2. *St. Petersburg*	4,233
3. *Tampa*	3,420
4. *Orlando*	2,743
5. *Jacksonville*	1,192

7. On July 1, 2019, Florida had a population of $21{,}477{,}737$. The area of Florida is $53{,}624.76$ square miles. Find the population density of Florida.
   ~ 401 people per square mile

For a kinesthetic approach, have cards listed with the city and the population density and allow students to do a quick card sort after doing the calculations.

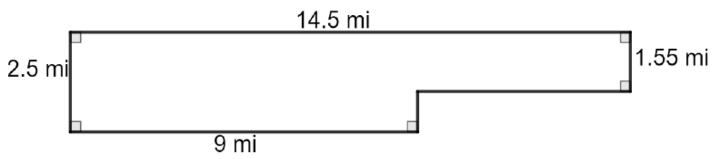
How does the population density of the state of Florida compare to the five largest cities in Florida? (The population density of the state as a whole is smaller than the population density of the five largest cities.)

9.2.3 Guided Instruction: Area and Density

Component Overview: *Students will begin working with contextual problems relating to area and density. Students will find the area and the population density in question 1 and notice a trend. Question 2 also requires students to find area and then use population density to determine the population of panthers in a certain area. Question 3 requires conversions of units to find population density.*

9.2.3 Guided Instruction: Area and Density

1. An outline of the city of Miramar, Florida, with measurements in miles (mi), is shown.



What strategy would be the best to use to find the area of the city of Miramar? (Divide the shape of the city into two separate rectangles and then add the areas together.)

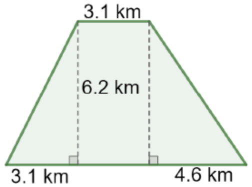
a. What is the area of the city, in square miles?
I made the shape into two rectangles.
Rectangle 1: $A = bh; A = 5.5\ \times 1.55 = 8.53\ mi^2$
Rectangle 2: $A = bh; A = 9\ \times 2.5 = 22.5\ mi^2$
$22.5 + 8.53 = 31.025\ mi^2$
The area of the city is $31.03 mi^2$

b. Determine the population density of Miramar for 1995 and 2015. Include units.

Year	Population	Population Density
1995	48,099	$1{,}550\ {}^{people}/_{mi^2}$
2015	136,996	$4{,}415\ {}^{people}/_{mi^2}$

c. How did the population density change between 1995 to 2015?
The population density almost tripled – meaning there are almost three times as many people on the same amount of land.

2. The Florida Fish and Wildlife Conservation Commission (FWC) tracks panthers in Florida. By tracking a panther, FWC can map the area in which a panther lives. The habitat of "Florida Panther 251" is shown with dimensions in kilometers, km.



a. What is the area, in square kilometers (km^2), of Florida Panther 251's habitat?

$A = \frac{1}{2}h(b_1 + b_2) = \frac{1}{2}6.2(3.1 + 10.8) = \frac{1}{2}6.2(13.9) = \frac{1}{2}86.18 = 43.09\ \mathrm{km}^2$

Students may calculate question 2a by either using the formula for the area of a trapezoid or by decomposing into rectangles and triangles and then adding together.

At one time, the entire southeastern United States was the habitat of the Florida panther. Today, their habitat is restricted to several national parks and wildlife preserves in southwest Florida. Today, the total combined area of the Florida panther habitat is $9{,}466\ km^2$.

b. Based on the area of Florida Panther 251's habitat, how many Florida panthers could be living in southwest Florida?

$9{,}466 \div 43.09 = 219.67$, *because you cannot have* $\frac{2}{3}$ *of a panther, I will round up. So* 220 *panthers could be living in southwest Florida.*

Monitor students on question 2b. Students may try to calculate population density instead of population. Also, students must round up their answers to the nearest whole number.

9.2.3 Guided Instruction: Area and Density (continued)

3. Town A has 20 city blocks, and each city block measures $\frac{1}{20}$ of a mile by $\frac{1}{2}$ mile. There are $6{,}500$ people in this town. Find the population density of the town.

   Area of 1 block: $A = bh = \frac{1}{20} \times \frac{1}{2} = \frac{1}{40} mi^2$

   Area of city blocks in Town A:
   $A = \left(\frac{1}{40} \times 20\right) = \frac{1}{2} mi^2$

   Population density: $\frac{6{,}500}{\frac{1}{2}} = 13{,}000$ *people / mi²*

What strategy would be the best to use to find the area of the town?

Consider having students draw a picture here to help better understand how to find the area of the town.

Provide additional scaffolding with students as needed when working with fractions to find the length and width of the town. Monitor that students are multiplying the fractions by 20.

9.2.4 Guided Practice: Density Based on Area

Component Overview: *Students will begin working with the population, area, and population density in more contextual questions. The problems in 9.2.4 Guided Practice are more complex than the other problems in the lesson. In question 2, students are given the population density and the persons per square mile, and it asks students to find the area of the city.*

9.2.4 Guided Practice: Density Based on Area

1. The number of leaves on a tree can be estimated using the crown of the tree. A tree crown is the top part of the tree where branches grow out from the main trunk. The width of the tree's crown is used as the diameter of the circle of the ground that lies under the tree's crown. The area where the leaves could fall is 4 times the area of the circle that lays under the tree's crown. The approximate number of leaves per square foot is based on the average size of a leaf for each type of tree. Use the table to approximate the number of leaves that could fall from each tree.

Tree	Crown (in feet)	Approximate Number of Leaves per Square Foot
Sycamore	35	5
Live Oak	16	14

Sycamore

Area of crown:

$$A_{crown} = \pi(17.5)^2 = 962.11$$

Area of leaves:

$$A_{leaves} = 4(962.11) = 3848.44$$

Number of Leaves:

$$5 = \frac{\#\ of\ leaves}{3848.44}$$

$$\#\ of\ leaves = 19242.2$$

Oak

Area of crown:

$$A_{crown} = \pi(8)^2 = 201.06$$

Area of leaves:

$$A_{leaves} = 4(201.06) = 804.24$$

Number of Leaves:

$$14 = \frac{\#\ of\ leaves}{804.24}$$

$$\#\ of\ leaves = 11259.36$$

2. The population density of Gainesville, Florida, is 2028.4 persons per square mile, and the population of Gainesville is $124{,}504$. What is the area, in square miles, of Gainesville?

Population density $= \frac{population}{area}$, *so area* $= \frac{population}{population\ density}$

using the terms given, this gives us an area of $61.38\ mi^2$.

Consider having students draw the two circles to represent the sycamore and live oak trees to have a visual representation of the question.

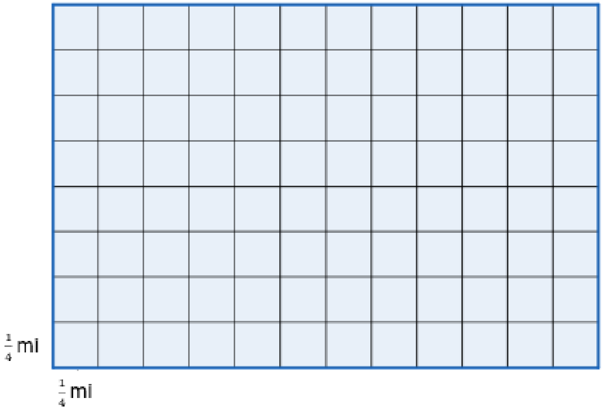
Once students have drawn visuals, prompt student thinking by asking the following questions below:

- *Does crown represent the diameter or the radius of the circle? (Diameter.)*
- *How do you find the radius of a circle when given diameter? (Divide by 2.)*
- *How do you find the number of leaves after you have found the area? (Multiply by the number of leaves per square foot.)*

Ask students how to find the area when given the population and people per square mile.

9.2.4 Guided Practice: Density Based on Area (continued)

3. Wildlife biologists divided a habitat in the Bear Island area of the Big Cypress National Preserve into 96 squares, as shown. Each square has the dimensions of $\frac{1}{4}$ mile (mi) by $\frac{1}{4}$ mi.



The biologists found that in the Bear Island habitat, there are 13.8 deer per square mile.

a. What is the population density for one of the $\frac{1}{4}$ mi by $\frac{1}{4}$ mi square?

There would be 16 of the $\frac{1}{4}$ mile squares within one square mile. Therefore, we would divide $13.8 \div 16 = 0.86$ *deer per square $\frac{1}{4}$ mile.*

b. How many deer live in the habitat area the biologists studied?

There are a total of 6 mi^2 and 13.8 deer per square mile. So, $6 \times 13.8 = 82.8 \approx 83$ *deer in the entire space.*

Consider having students use colored pencils and/or pens to divide and label each section on the grid.

9.2.5 Wrap-Up: Cool Down

Component Overview: Students write a short summary of their learning in the lesson. Student responses can be used to identify misconceptions and plan for upcoming instruction.

9.2.5 Wrap-Up: Cool Down

1. Write an explanation to a classmate who might have missed today's lesson, in your own words, about population density.

   Answers will vary.

Consider giving some extra support with the sentence stems given. Encourage students to be as specific as possible here for self-reflection.